

Describing Meaning — Week 5

Determiners and Generalised Quantifiers

Notation

We treat common nouns and verb phrases as denoting sets of individuals. Let A and B be sets. We will write:

- A for the set denoted by the restrictor (e.g. *hippos*)
- B for the set denoted by the nuclear scope (e.g. *swim*)
- $\text{DET}(A)(B)$ for the truth-conditions of a determiner combining with its two set arguments.

Two determiners

- **Every:** $\text{EVERY}(A)(B)$ is true iff $A \subseteq B$.
- **Some:** $\text{SOME}(A)(B)$ is true iff $A \cap B \neq \emptyset$.

We will also consider some **fictional** determiners later in the handout, including one that corresponds to the English description “between 3 and 5”.

Exercise 1

Let:

$$A = \{a, b, c, d\} \quad B = \{b, c, e\}.$$

1. Compute $A \cap B$ and $A \setminus B$.
2. Is $A \subseteq B$? Is $B \subseteq A$?
3. Is $\text{EVERY}(A)(B)$ true?
4. Is $\text{SOME}(A)(B)$ true?

Exercise 2: derive basic determiner meanings

Using only $\subseteq$, $\cap$, $\emptyset$, $|\cdot|$, and set difference $\setminus$, give set-theoretic truth conditions for each determiner below.

1. $\text{NO}(A)(B)$
2. $\text{ATLEAST2}(A)(B)$
3. $\text{MOST}(A)(B)$

Fictional determiners

Below are four possible determiner meanings stated informally in English.

- **ex2:** “Exactly two A ’s are B .”
- **equi:** “The number of A ’s equals the number of B ’s.”
- **even:** “An even number of A ’s are B .”
- **between3and5:** “Between 3 and 5 A ’s are B .”

Exercise 3

Give set-theoretic truth conditions for:

1. $\text{EX2}(A)(B)$
2. $\text{EQUI}(A)(B)$
3. $\text{EVEN}(A)(B)$
4. $\text{BETWEEN3AND5}(A)(B)$

Exercise 4

Let:

$$A = \{a, b, c\} \quad B = \{b, c, d, e\}.$$

1. Is $\text{EX2}(A)(B)$ true?
2. Is $\text{EQUI}(A)(B)$ true?
3. Is $\text{EVEN}(A)(B)$ true?
4. Is $\text{BETWEEN3AND5}(A)(B)$ true?

Conservativity

A widely discussed hypothesis is that natural language determiners obey a strong constraint called **conservativity**. Informally, conservativity says:

To evaluate a sentence of the form *DET A are B*, you only need to look at which *A*'s are *B*. It should never matter whether things *outside* of *A* are in *B*.

This is often discussed as a candidate **semantic universal** about determiners cross-linguistically: while languages vary enormously in the inventory of determiners they lexicalise, determiners appear to be constrained in systematic ways, and conservativity is one of the most robust such constraints.

Formally, a determiner DET is **conservative** iff:

$$\text{DET}(A)(B) \iff \text{DET}(A)(A \cap B).$$

Example

$$\begin{aligned} \text{EVERY}(A)(B) \text{ is true iff } A \subseteq B. \\ \text{EVERY}(A)(B) &\iff \text{EVERY}(A)(A \cap B), \\ A \subseteq B &\iff A \subseteq (A \cap B). \end{aligned}$$

Exercise 5: conservativity

For each determiner below, determine whether it is conservative.

1. SOME
2. NO
3. MOST
4. ATLEAST2
5. EX2
6. EQUI
7. EVEN
8. BETWEEN3AND5

Monotonicity

Another family of constraints concerns how determiners behave when we replace the nuclear scope predicate with a *stronger* or *weaker* one.

A predicate is **stronger** if it applies to fewer things (a smaller set), and **weaker** if it applies to more things (a larger set). So if $B \subseteq C$, then *B* is stronger than *C*, and *C* is weaker than *B*.

- A determiner is **upward monotone** in its second argument if, whenever it holds with a stronger predicate *B*, it also holds with any weaker predicate *C* that contains it.

- A determiner is **downward monotone** in its second argument if, whenever it holds with a weaker predicate C , it also holds with any stronger predicate B inside it.

Formally:

- **Upward monotone:** if $B \subseteq C$, then $\text{DET}(A)(B) \Rightarrow \text{DET}(A)(C)$.
- **Downward monotone:** if $B \subseteq C$, then $\text{DET}(A)(C) \Rightarrow \text{DET}(A)(B)$.

Example

Let $B = \text{is a mammal}$, $C = \text{is an animal}$. Since every mammal is an animal, we have $B \subseteq C$.

If every platypus is a mammal, then every platypus is an animal.

So *every* is upwards monotone

However,

If every platypus is an animal, then every platypus is a mammal.

Is not valid, so *every* is **not** downwards monotone

Exercise 6

For each determiner below, determine whether it is upward monotone in B , downward monotone in B , or neither.

1. SOME
2. NO
3. MOST
4. ATLEAST2
5. EX2
6. EQUI
7. EVEN
8. BETWEEN3AND5